



FURTHER MATHEMATICS UNIT 3 APPLICATION TASK SAC 1 2019

CORE – DATA ANALYSIS : BUYING A CAR

VCAA NUMBER: SOLUTIONS **TEACHER:** JGA

Dates: Wednesday 24th April 3:30-5:40PM (all students)

Plus 3 periods in class (1 double and 1 single) in the following few days: Friday 26th April to Tuesday 30th April

Duration: 10 minutes reading time plus 2 hours plus 3 x 50 minutes.

Materials allowed: 1 x Bound reference, CAS calculator, scientific calculator, ruler, pens/pencils

Outline of SAC

Students will be given PART ONE of the SAC on Wednesday 24th April and will have 10 minutes reading time plus 2 hours on Monday 23rd April. They will be given PART TWO to complete in a single lesson on Friday 26th April and PART THREE to complete in a double lesson Monday 29th or Tuesday 30th April.

At the end of each session the student must:

1. Submit their SAC,
2. Submit their bound reference,
3. Submit their CAS calculator
4. Sign to state that they have submitted all these items and that the work is their own.

Students will be given their SAC, bound reference and CAS back at the start of each session.

Total Marks: 83

PART ONE

Question 1 : 16mks Question 2 : 13mks

PART TWO

Question 3 : 17mks

PART THREE

Question 4 : 17mks Question 5: 20mks

In order to satisfactorily complete this SAC students must demonstrate an understanding of the following three outcomes. These outcomes are covered throughout the SAC.

The following table summarises what each outcome entails in terms of key knowledge and skills. Please note that not all points are covered in this SAC- indicated by an NA (not assessed).

Outcome 1 (10 marks): On completion of this unit the student should be able to define and explain key concepts and apply related mathematical techniques and models as specified in Area of Study 1 in routine contexts. To achieve this outcome the student will draw on knowledge and skills outlined in Area of Study 1.	
Key knowledge	Satisfactory
• types of data: categorical (nominal and ordinal) and numerical (discrete and continuous)	
• frequency tables, bar charts including segmented barcharts, histograms, stem plots, dot plots, and their application in the context of displaying and describing distributions	
• log (base 10) scales, and their purpose and application	
• five-number summary and boxplots (including the designation and display of possible outliers)	
• mean and standard deviation	
• normal model and the 68–95–99.7% rule, and standardised values (z-scores)	
• response and explanatory variables	
• two-way frequency tables, segmented bar charts, back-to-back stem plots, parallel boxplots, and scatterplots, and their application in the context of identifying and describing associations	
• correlation coefficient, r , its interpretation, the issue of correlation and cause and effect	
• least squares line and its use in modelling linear associations	
• data transformation and its purpose	
• time series data and its analysis	NA
Key skills	
• construct frequency tables and bar charts and use them to describe and interpret the distributions of categorical variables	
• answer statistical questions that require a knowledge of the distribution/s of one or more categorical variables	
• construct stem and dot plots, boxplots, histograms and appropriate summary statistics and use them to describe and interpret the distributions of numerical variables	
• answer statistical questions that require a knowledge of the distribution/s of one or more numerical variables	
• solve problems using the z-scores and the 68–95–99.7% rule	
• construct two-way tables and use them to identify and describe associations between two categorical variables	NA
• construct parallel boxplots and use them to identify and describe associations between a numerical variable and a categorical variable	
• construct scatterplots and use them to identify and describe associations between two numerical variables	
• calculate the correlation coefficient, r , and interpret it in the context of the data	
• answer statistical questions that require a knowledge of the associations between pairs of variables	
• determine the equation of the least squares line giving the coefficients correct to a required number of decimal places or significant figures as specified	
• distinguish between correlation and causation	
• use the least squares line of best fit to model and analyse the linear association between two numerical variables and interpret the model in the context of the association being modelled	
• calculate the coefficient of determination, r^2 , and interpret in the context of the association being modelled and use the model to make predictions, being aware of the problem of extrapolation	
• construct a residual analysis to test the assumption of linearity and, in the case of clear non-linearity, transform the data to achieve linearity and repeat the modelling process using the transformed data	
• identify key qualitative features of a time series plot including trend (using smoothing if necessary), seasonality, irregular fluctuations and outliers, and interpret these in the context of the data	NA
• calculate, interpret and apply seasonal indices	NA
• model linear trends using the least squares line of best fit, interpret the model in the context of the trend being modelled, use the model to make forecasts being aware of the limitations of extending forecasts too far into the future	

Outcome 2 (20 marks): On completion of this unit the student should be able to select and apply the mathematical concepts, models and techniques as specified in Area of Study 1 in a range of contexts of increasing complexity. To achieve this outcome the student will draw on knowledge and skills outlined in Area of Study 1			
Key knowledge			
- the facts, concepts and techniques associated with data analysis - standard models studied in data analysis and their area of application - general formulation of the concepts, techniques and models studied in data analysis - assumptions and conditions underlying the use of the concepts, techniques, and models associated with data analysis			
Key skills			
- identify, recall and select facts, concepts, models and techniques needed to investigate and analyse statistical features of a data set with several variables that can include time series data - interpret and report the results of a statistical investigation or of completing a modelling or problem-solving task in terms of the context under consideration, including discussing the assumptions in application of these models			
Outcome 3 (10 marks): On completion of this unit the student should be able to select and appropriately use numerical, graphical, symbolic and statistical functionalities of technology to develop mathematical ideas, produce results and carry out analysis in situations requiring problem-solving, modelling or investigative techniques or approaches. To achieve this outcome the student will draw on knowledge and related skills outlined in Area of Study 1.			
Key knowledge			
- the difference between exact numerical and approximate numerical answers when using technology to perform - computation, and rounding to a given number of decimal places or significant figures - domain and range requirements for specification of graphs of models and relations, when using technology - the relation between numerical, graphical and symbolic forms of information about models and equations and - the corresponding features of those functions and equations - similarities and differences between formal mathematical expressions and their representation by technology - the selection of an appropriate functionality of technology in a variety of mathematical contexts			
Key skills			
- distinguish between exact and approximate presentations of mathematical results produced by technology, and interpret these results to a specified degree of accuracy in terms of a given number of decimal places or significant figures - use technology to carry out numerical, graphical and symbolic computation as applicable - produce tables of values, families of graphs and collections of other results using technology, which support general analysis in problem-solving, investigative and modelling contexts - identify the relation between numerical, graphical and symbolic forms of information about models and equations and the corresponding features of those models and equations - select an appropriate functionality of technology in a variety of mathematical contexts, related to data analysis, and provide a rationale for these selections - relate the results from a particular technology application to the nature of a particular mathematical task (investigative, problem solving or modelling) and verify these results - specify the process used to develop a solution to a problem using technology, and communicate the key stages of mathematical reasoning (formulation, solution, interpretation) used in this process			
Overall result:	Outcome 1	Outcome 2	Outcome 3

After SACs have been marked, teachers will decide if a student has demonstrated enough knowledge throughout their SAC to be given a satisfactory. If a student can't demonstrate through her SAC that she understands the content covered then the student must demonstrate her knowledge by either: 1. Showing her teacher her up-to-date workbook; and/or, 2. Completing a short test.

SAC

INTRODUCTION:

New cars are safer to drive, but they are expensive to buy. This SAC explores the prices of cars of various ages, various sizes and various makes and attempts to find a satisfactory relationship between the age of a car and its price.

Ms G would like her son Rourke to buy a recent model car because of the additional safety features. She decides to investigate 2016 models of cars on carsales.com.au and collates the average price of cars that are for sale in March. All vehicles are standard automatic petrol sedans.

Ms G is interested in a few different research questions.

1. Is there a difference between sedan size and reaction price?
2. Do some cars lose value over time more quickly than others?
3. Is it possible to predict a car's price from its age?

These questions and others will be investigated in this assessment.

PART ONE

2016 Car Prices

Car Prices		
Car Type	Sedan Size	2016 Price (\$)
Ford Focus	Small	18 308
Honda Civic	Small	20 216
Holden Cruze	Small	15 530
Hyundai Elantra	Small	18 090
Kia Cerato	Small	15 674
Mazda 3	Small	19 882
Maserati Ghibli	Small	95 518
Mitsubishi Lancer	Small	16 302
Nissan Pulsar	Small	14 730
Subaru Impreza	Small	18 499
Toyota Corolla	Small	16 702
Audi A4	Medium	55 282
BMW M3	Medium	103 043
Holden Malibu	Medium	14 310
Honda Accord	Medium	29 022
Hyundai Sonata	Medium	18 800
Jaguar XE	Medium	56 504
Kia Optima	Medium	22 367
Mazda 6	Medium	29 071
Mercedes Benz C63	Medium	108 813
Nissan Altima	Medium	17 404
Subaru Liberty	Medium	24 470
Toyota Camry	Medium	19 709
Toyota Prius	Medium	24 554

Question 1 : Analysing the 2016 data. (16 marks)

- a. State whether the following variables are numerical (discrete or continuous) or categorical (nominal or ordinal). Two words are needed to describe each variable.

Outcome

- i. Sedan Size

Categorical ordinal

(1)

- ii. Car Type

Categorical nominal

(1)

- iii. Price

Numerical continuous

(1)

(3 marks)

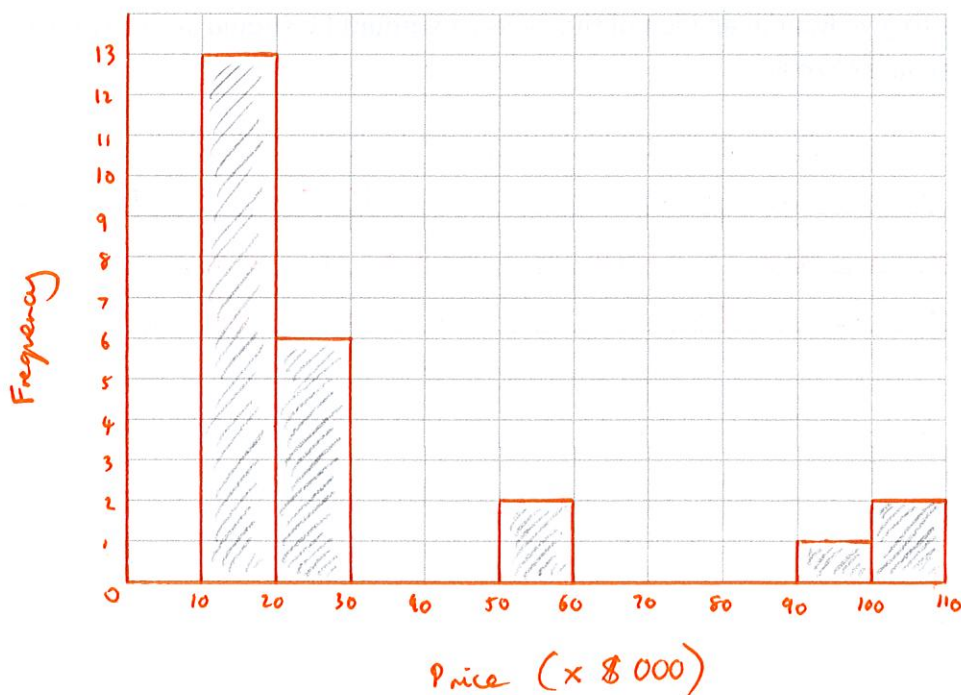
- b. Complete the frequency table below for 2016 car prices for all cars.

Price	Frequency	Interval	Frequency
\$10 000- < \$20 000	### ## III 13	\$60 000- < \$70 000	0
\$20 000- < \$30 000	### I 6	\$70 000- < \$80 000	0
\$30 000- < \$40 000	0	\$80 000- < \$90 000	0
\$40 000- < \$50 000	0	\$90 000- < \$100 000	1
\$50 000- < \$60 000	II 2	\$100 000- < \$110 000	2

(2 marks)

- c. Construct an appropriate graph on the grid below to summarize the 2016 car data.

(-1 for error)



(1) Scale
(1) Axes
(1) Appropriate Graph

(3 marks)

d. Using your graph from part c., describe the 2016 Car Price data in terms of shape, centre and spread.

② Positively skewed with outliers (or possible outliers) (1)

② Median interval \$10 000 - \$20 000 (1)

③ Range 110 000 - 10 000 = \$100 000 (1)

(3 marks)

e. Ms G realised that she may have included Luxury cars in her search. Perform an outlier test and state the Luxury cars in the 2016 data.

$$Q_1 = 17053 \quad \text{Med} = 19795.5 \quad Q_3 = 29046.5 \quad (1)$$

③

$$\begin{aligned} \text{Upper Fence} &= Q_3 + 1.5 \times \text{IQR} \\ &= 29046.5 + 1.5(29046.5 - 17053) \\ &= 47036.75 \end{aligned} \quad (1)$$

$\therefore$ 5 outliers above upper fence

They are Audi A4, BMW M3, Mercedes Benz C63, Maserati Ghibli, Jaguar XE (1)

(3 marks)

f. Ms G would like to give Rourke an idea of the average amount he should save to buy a car. Should she use the original data? Explain.

No (1)

② Mean would not give a good indication of the centre of the data as the 5 outliers would increase the mean a great deal. Median should be used in this case. (1)

(2 marks)

Question 2 : Refining the search. (13 marks)

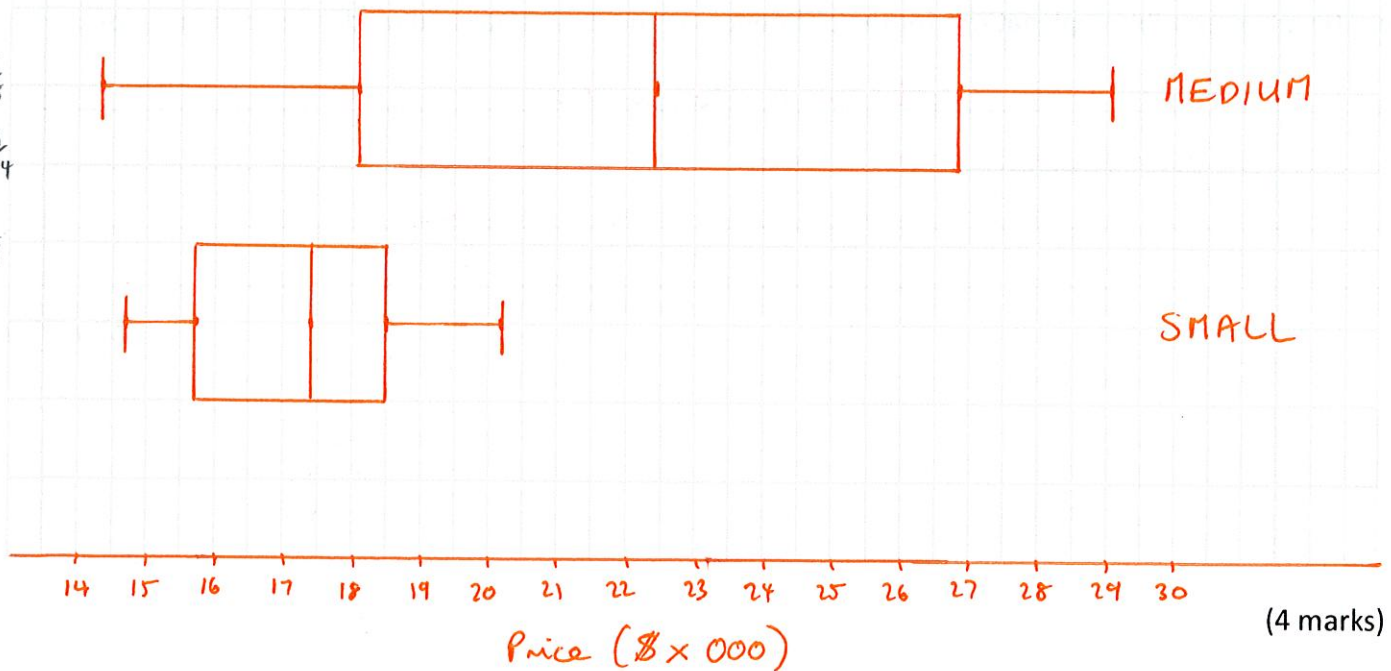
- a. Ms G decides to take out the luxury cars from her 2016 search and investigate the difference in price of small sedans compared to medium sedans. Calculate the five number summaries (to the nearest dollar) for the prices of cars for each of the sedan sizes.

(3)

	Small	Medium
min	14730	14310
Q_1	15674	18102
med	17396	22367
Q_3	18499	26788
max	20216	29071

(2 marks)

- b. Construct parallel boxplots for the data for 2016 small and medium non-luxury sedans.

(1) $\times \frac{5}{8}$ (2) $\times \frac{1}{4}$ (3) $\times \frac{1}{8}$ 

(1) Scale

(1) Labels

(2) Boxplot $-\frac{1}{2}$ for each error

- c. Comparing the shape, centre and spread of the boxplots, compose a summary using statistics for Ms G to give to Rourke regarding the price of 2016 small non-luxury sedans compared to medium non-luxury sedans.

Shape - Both symmetric

Centre - Median much higher for medium cars (\$22367)
compared to small cars (\$17396)

Spread - Range much higher for medium cars (\$14761)
compared to small cars (\$5486)

or IQR higher for medium (\$8684) compared to small (\$2825)

(3 marks)

Using the parallel boxplots you constructed, answer question d.

- d. Approximately what percentage of 2016 small sedans and medium sedans were above \$18 000?

Small - over 25% (accept up to 35%)

Medium - 75%

(2 marks)

- e. Give two possible explanations for the differences in the 2016 small and medium sedan box plots.

Larger cars cost more to make.

Larger cars may have more features.

Medium cars may have larger engines, etc.

(2 marks)

PART TWO

Question 3 : Which cars hold their value the most over time? (17 marks)

Rourke has decided he can only afford a small sedan. Ms G wants him to make a good investment so she was wondering if some cars decreased in value faster than others. She explored the current 2019 prices of the same vehicles and this data can be found in table 2. She omitted the luxury cars and the Holden and Nissan were not produced in 2019.

Car	Car Type	2016 Price (\$)	2019 Price (\$)
1	Ford Focus	18 308 $\bar{z}$	24 390 $\bar{z}$
2	Honda Civic	20 216 $\bar{z}$	24 390 $\bar{z}$
3	Hyundai Elantra	18 090 $\bar{z}$	23 790 $\bar{z}$
4	Kia Cerato	15 674 $\bar{z}$	22 290 $\bar{z}$
5	Mazda 3	19 882 $\bar{z}$	25 490 $\bar{z}$
6	Mitsubishi Lancer	16 302 $\bar{z}$	21 000 $\bar{z}$
7	Subaru Impreza	18 499 $\bar{z}$	22 890 $\bar{z}$
8	Toyota Corolla	16 702 $\bar{z}$	24 370 $\bar{z}$

- a. Calculate the mean and standard deviation for the non-luxury 2016 data and for the 2019 data found in the table above. Place the values in the table below to 2 significant figures.

	2016	2019
Mean	$\frac{17959.125}{18\ 000}$	$\frac{23576.25}{24\ 000}$
Standard Deviation	$\frac{1637.48}{16.00}$	$\frac{1435.52}{1400}$

(2 marks)

- b. Is the mean an appropriate measure of centre for these car prices? Explain why or why not.

Yes

because there are no outliers and distribution is approximately symmetrical.

(2 marks)

- c. Ms G is going to use the 68-95-99.7% rule to analyse the 2016/2019 data. Explain why the 68-95-99.7% rule should or should not be used with this data.

This rule should be used as the distribution appears to be normally distributed.

(1 mark)

- d. Using CAS, randomly select one of the cars from the above 2016/2019 data. State below the car and the method that you used on CAS to select this vehicle.

③ Give each a number from 1-8 and generate a random number
by CAS - Menu $\rightarrow$ 5: Probability $\rightarrow$ 4: Random $\rightarrow$ 2: Integer
randint(1,8) or randint(1,8,1)

(1 mark)

- e. Using your results from a., calculate the standardised score for your selected vehicle for the 2019 data correct to 2 decimal places.

① $z = \frac{x - \bar{x}}{s}$ See page 11.

(2 marks)

- f. For the standardised score obtained in part e. above, determine the approximate percentage (or range) for the number of cars below this value and explain your reasoning.

② See pages 12a, 12b

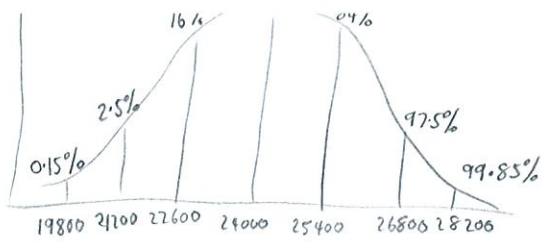
(2 marks)

- g. Using your results from a., calculate the standardised score for the same vehicle for the 2016 car data correct to 2 decimal places.

① See page 11.

(2 marks)

2019

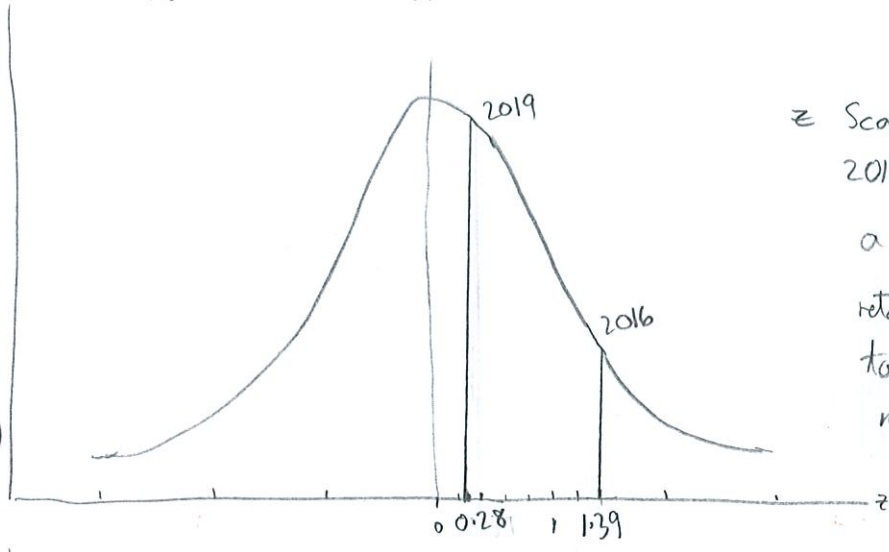


Car 2

Honda Civic

2019 $z = 0.28$

$\approx 50\%$ ($\leq 60\%$) below
 $< \$24000$ (< 25000)



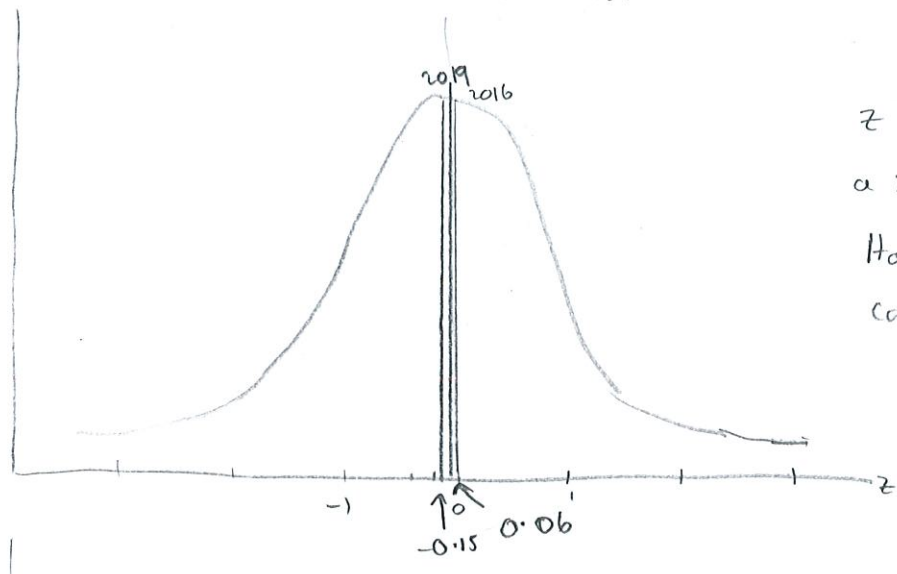
z Score increased for 2016 so this is a good buy as retains value compared to the rest of the market. A good investment.

Car 3

Hyundai Elantra

2019 $z = -0.15$

$\approx 50\%$ ($\geq 40\%$) below
 $< \$24000$ (> 23000)



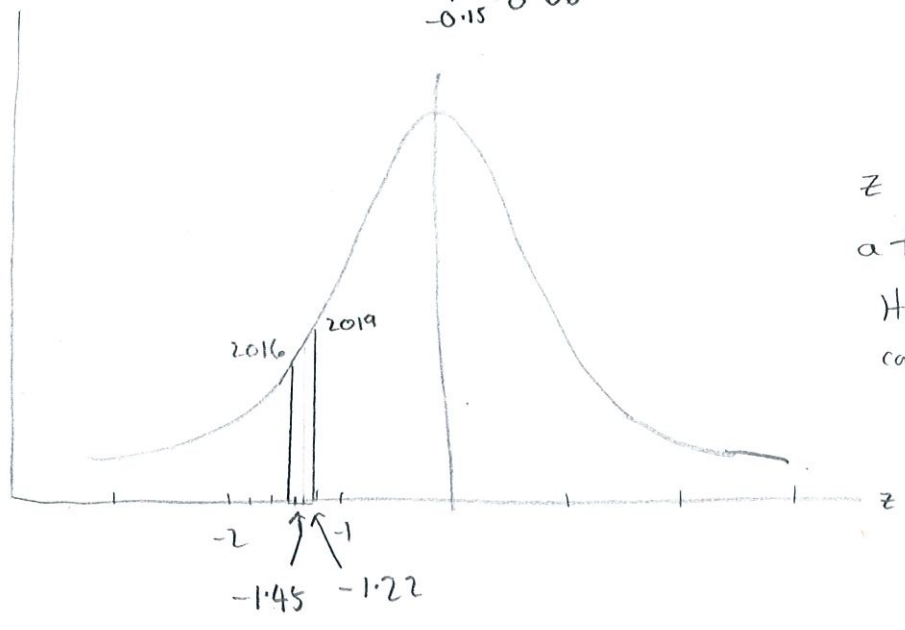
z score increased a tiny bit. Holds its value compared to rest.

Car 4

Kia Cerato

2019 $z = -1.22$

$\approx 16\%$ below ($\geq 10\%$)
 $< \$22600$ (> 21500)



z score decreased a tiny bit. Holds its value compared to rest.

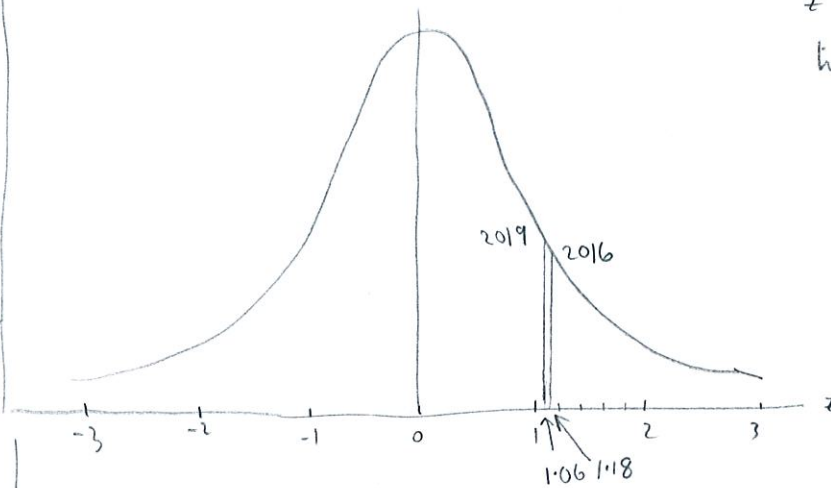
Car 5

Mayda 3

2019 $z = 1.06$

$\approx 84\%$ ($>88\%$) below

$< \$25400$ (26000)



z score improved a tiny bit. Retained comparative value.

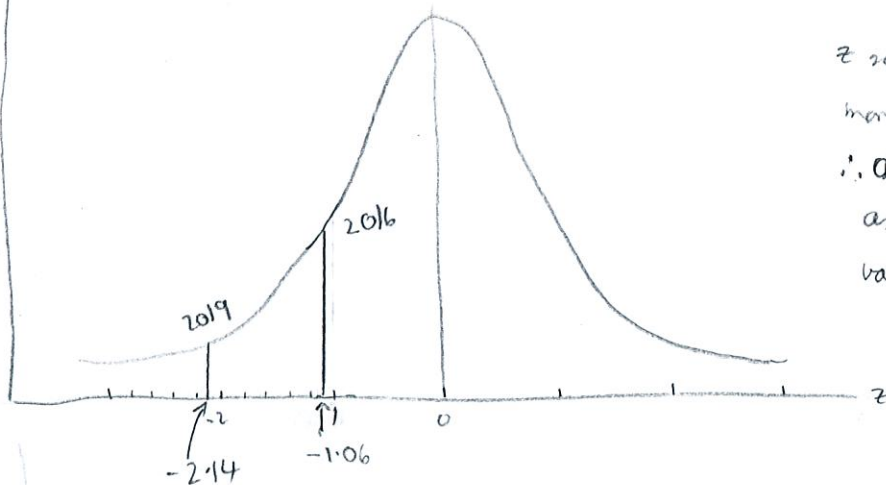
Car 6

Mitsubishi Lancer

2019 $z = -2.14$

$\approx 2.5\%$ (>2.3) below

$< \$21200$ (20500)



z score increased by more than one s.d.

$\therefore$ a good buy as increases value comparatively

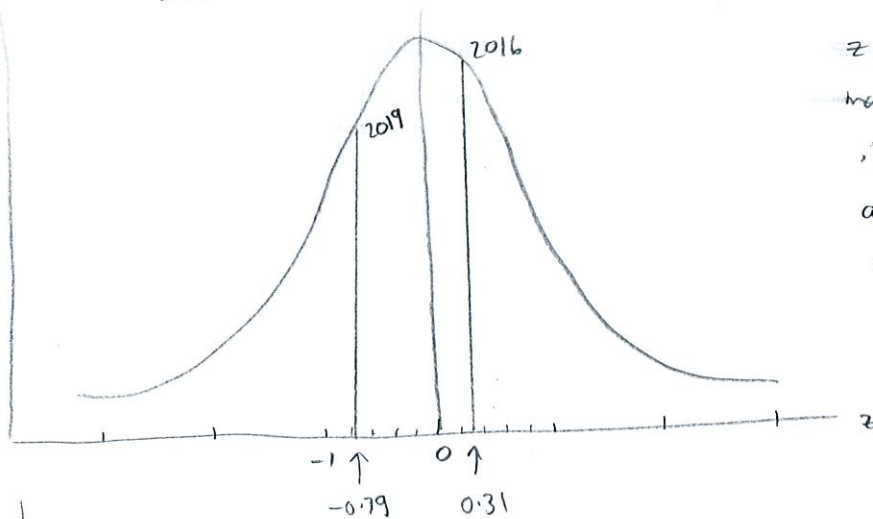
Car 7

Suzuki Impreza

2019 $z = -0.79$

$\approx 16\%$ (<25) below

$< \$22600$ (23500)



z score increased by more than one s.d.

$\therefore$ a good buy as increases value comparatively

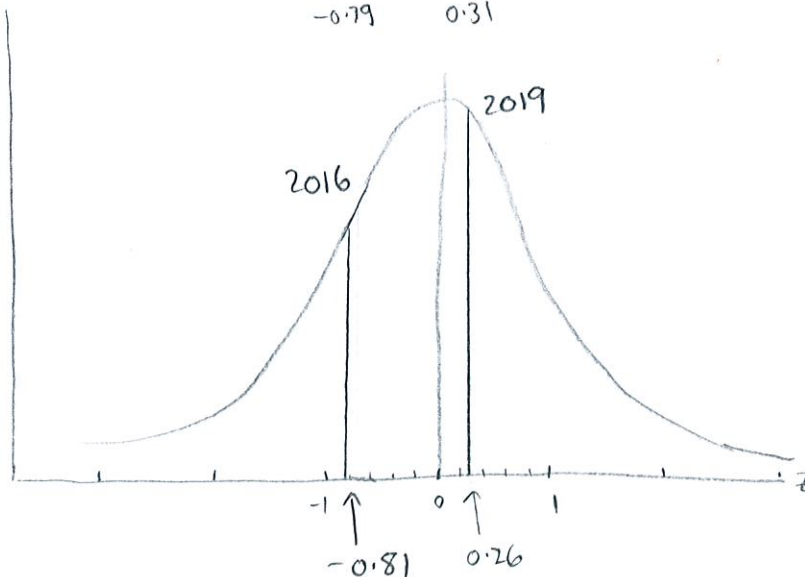
Car 8

Toyota Corolla

2019 $z = 0.26$

$\approx 50\%$ (60%) below

$< \$24000$ (25000)



z score decreased by more than one s.d.

$\therefore$ not a good buy as decreases value compared to others.

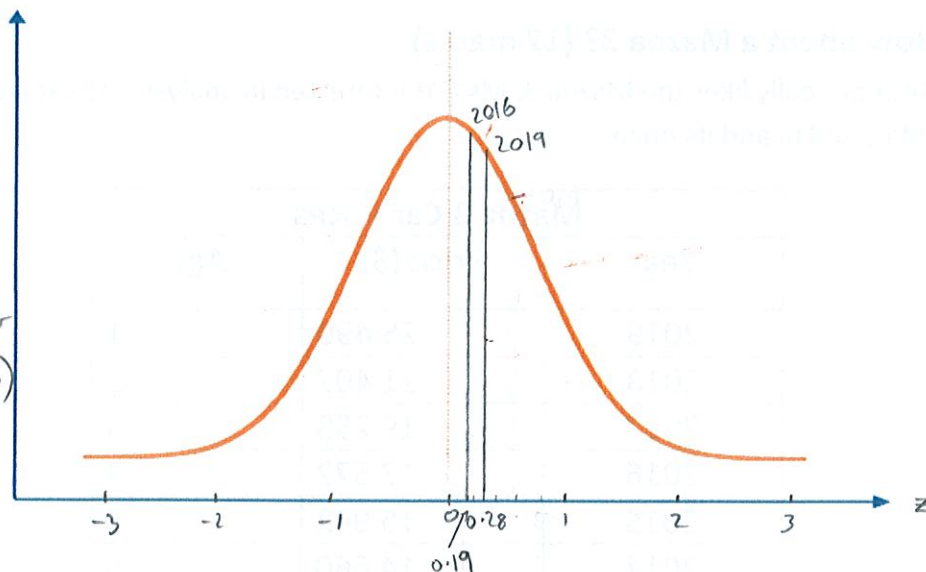
- h. On the normal graph below, approximately place your values found in parts e. and g. and label them. Compare and comment on what the 2 standard scores imply for your chosen car.

Car 1

Ford Focus

②

2019 $z = 0.28$
 $\approx 50\%$ ($\leq 60\%$) value
 $< \$24,000$ ($< 25,000$)



z score decreased slightly for 2016.
 Holds its value compared to the rest.
 See pages 12a, 12b.

(2 marks)

- i. Rourke is thinking of buying a car in the top 16% price range. Using your results from a., determine the minimum amount that he would spend on a 2016 car and a 2019 car.

②

$$2016 \quad 18000 + 1800 = \$19800 \quad (19596.61)$$

③

$$2019 \quad 24000 + 1400 = \$25400 \quad (25011.77)$$

(2 marks)

- j. What is the actual percentage of 2016 cars that fall in the top 16% price range?

③

$$\frac{2}{8} \times 100 = 25\%$$

(1 mark)

PART THREE

Question 4 : How about a Mazda 3? (17 marks)

Rourke decides that he really likes the Mazda 3. Ms G is interested in analysing the relationship between the age of a Mazda 3 sedan and its price.

Mazda 3 Car Prices		
Year	Price (\$)	Age
2019	25 490	1
2018	21 407	2
2017	19 258	3
2016	17 572	4
2015	15 942	5
2014	14 660	6
2013	12 890	7
2012	11 725	8
2011	11 230	9
2010	9 526	10
2009	8 733	11
2008	6 062	12
2007	7 660	13
2006	6 332	14
2005	6 180	15
2004	4 925	16

- a. In this case, what is the response variable?

①

Price

(1 mark)

- b. Use CAS to plot the above points and describe the relationship between the age of the Mazda 3 and the price in dollars in terms of strength, direction and form.

③

Strong, negative, non-linear

②

②

(3 marks)

- c. The value of the coefficient of determination for the least squares relationship between the price and the age is 0.9397, correct to 4 decimal places. Interpret the meaning of this value in terms of the given data.

About 93.97% of the variation in the price of the Mazda3 can be explained by the variation in the age of the car.

(1 mark)

- d. ^{Using the information from c. above,} Determine the value of Pearson's correlation coefficient for this data correct to 4 decimal places.

$$-0.969381 \dots \approx -0.9694$$

(1 mark)

- e. Given the following information and also using your answer to part d., calculate the slope and the y-intercept for the equation of the least squares regression line for this data. Show your working. Then write the equation using correct variables and the coefficients correct to 3 significant figures.

mean age of car = 8.5

mean price of car = \$12 500

standard deviation of car ages = 4.76

standard deviation of car prices = \$6 100

$$b = \frac{r s_y}{s_x} = \frac{-0.9694 \times 6100}{4.76} = -1242.298 \approx -1240$$

$$a = \bar{y} - b\bar{x} = 12500 - (-1240 \times 8.5) = 23040 \approx 23000 \quad (\text{accurate } 23100) \quad 1960(1940)$$

$$\text{Price} = 23000 - 1240 \times \text{age}$$

(3 marks)

- f. Explain the meaning in real terms of the slope of this line.

For each increase of one year's age (as the car gets one year older), the price decreases by approximately \$1240.

(1 mark)

- g. Use the least squares regression equation to calculate the price of a 5-year-old Mazda 3.

$$\text{Price} = 23000 - 1240 \times 5 = \$16800$$

(1 mark)

- h. Rourke is looking at a 5 year old Mazda 3 which is advertised at \$14 999. Calculate the residual of the price for this vehicle. Explain what this means to Rourke.

$$\begin{aligned}\text{residual} &= \text{actual} - \text{predicted} \\ &= 14999 - 16800 \\ &= -1801\end{aligned}$$

The car is \$1801 below the predicted price $\therefore$ it is a good deal.

(2 marks)

- i. Use CAS to perform a linear regression and check your answer to e. is close. Complete the following table of residuals correct to the nearest dollar for the Mazda 3 data.

Age of Mazda 3	Residual of price (\$)
1	3 671
2	834
3	-69
4	-509
5	-893
6	-929
7	-1453
8	-1372
9	-622
10	-1 080
11	-627
12	-2 052
13	792
14	710
15	1 804
16	1 795

(2 marks)

- j. Use CAS to see the residual plot. What does the residual plot indicate about the relationship between age of a Mazda 3 and its price?

The shape is 

Not a random residual plot. Large residuals 3671 to -2052
 $\therefore$ Not a linear relationship.

(2 marks)

Question 5 : Maybe a transformation is needed? (20 marks)

Ms G has come to the conclusion that a transformation may enable her to make a more accurate prediction of the price of the Mazda 3 given its age. Use the table at the start of Question 4 to help you answer the following questions.

- a. Perform each of the transformations listed below and write the value of r^2 and the coefficients in the equations correct to 2 significant figures. Describe the residual plot of each transformation.

Relationship	r^2 value	Equation	Residual plot description
Price vs. age	0.94	$\text{price} = 23000 - 1200 \times \text{age}$	Not random. Large residuals.
Price vs. $\log(\text{age})$	0.97	$\text{price} = 27000 - 18000 \times \log(\text{age})$	Not random. Large residuals. 1200 $\rightarrow$ -2000
Price vs. $1/\text{age}$	0.74	$\text{price} = 7900 + 22000 \times \frac{1}{\text{age}}$	Not random. Large residuals, 4000 $\rightarrow$ -4000
$\log(\text{price})$ vs. age	0.98	$\log(\text{price}) = 4.4 - 0.045 \times \text{age}$	Random. Small residuals. 0.06 $\rightarrow$ -0.08
$1/\text{price}$ vs. age	0.92	$\frac{1}{\text{price}} = 0.000016 + (1.0 \times) \text{age}$	Not random. Tiny residuals, 0.00003 $\rightarrow$ -0.00002

(5 marks)

- b. State which relationship you consider best and give reasons for your answer.

$\log(\text{price})$ vs age (1)
as random residual plot, high r^2 value and small residuals. (1)

(2 marks)

- c. Using your equation from part a. above, predict the price of a 5-year-old Mazda 3 using the relationship that you consider best correct to the nearest dollar.

$$\begin{aligned} \log \text{ price} &= 4.4 - 0.045 \times \text{age} \\ \log \text{ price} &= 4.4 - 0.045 \times 5 \\ \text{price} &= 14962.356 \dots \approx \$14962 \end{aligned}$$

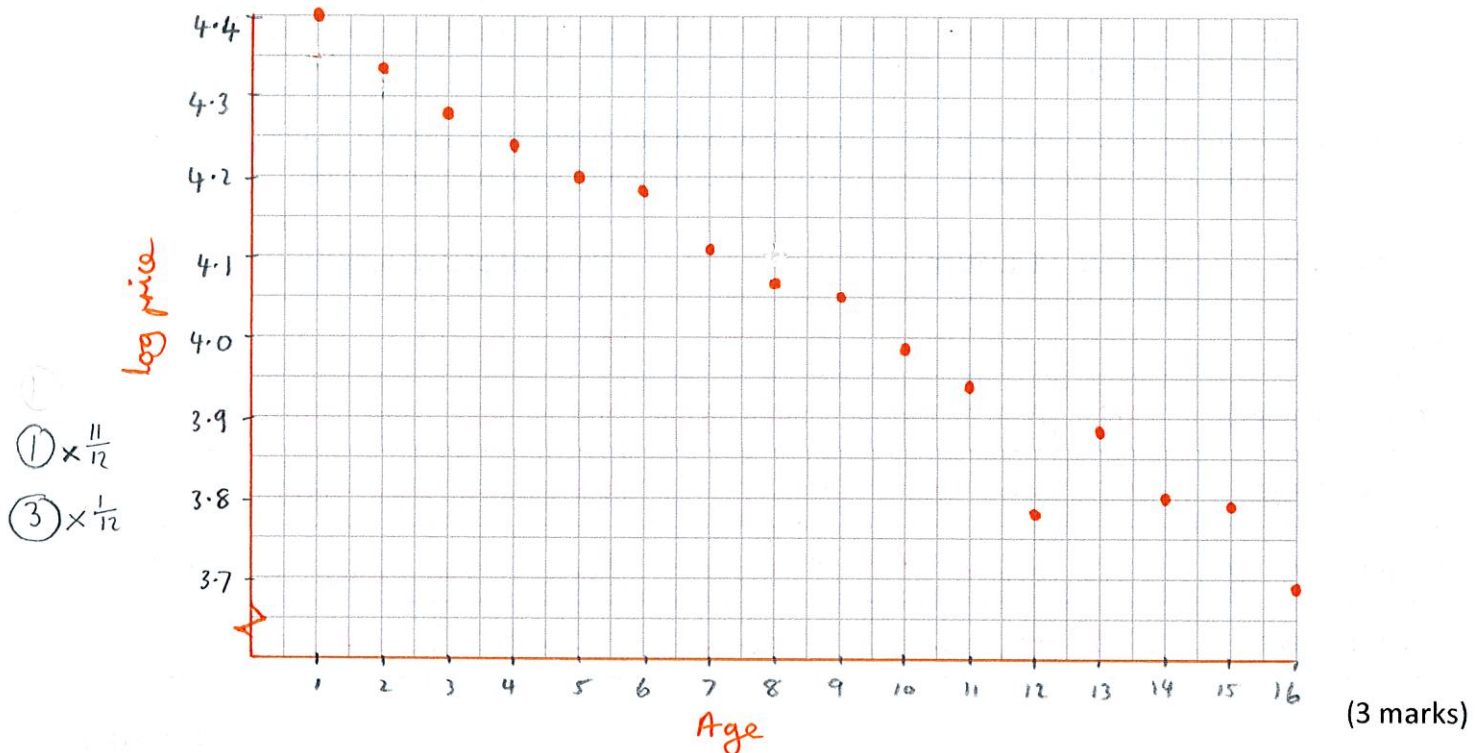
(2 marks)

- d. Comment on the reliability of your prediction in part c.

This is a reliable prediction as (1)
within the data range. This is interpolation. (1)

(2 marks)

- e. Construct a graph of the relationship that you consider best, showing the scatterplot of the relationship **after** transformation.



- f. Does the scatterplot after transformation indicate that the transformation was successful? Explain your answer.

Yes.

This scatterplot is more linear.

(1)

(1)

(2 marks)

- g. Calculate the residual for the 5 year old Mazda 3 advertised at \$14 999.

$$\begin{aligned} \text{Residual} &= \text{actual} - \text{predicted} \\ &= 14999 - 14962 \\ &= 37 \end{aligned}$$

Concept from C

(2 marks)

- h. Compare the residual you obtained in g. above to the residual you obtained in Question 4 part h. Explain, referring to the result of the transformation, **why** any difference was observed.

Using the log price transformation, the residual was much smaller and the actual price was almost the same as the predicted. This is because there is a better linear relationship using the transformed data.

(2 marks)

END OF SAC